G. NARAYANAMMA INSTITUTE OF TECHNOLOGY AND SCIENCE

(AUTONOMOUS) (For Women)

Shaikpet, Hyderabad - 500 104

Department of Computer Science and Engineering

B. Tech Minor in Data Science

III B. Tech. I Semester Academic Year 2026-27

Assignment-1

SUBJECT: Introduction to Data Science

SUBJECT CODE:---- **Max Marks: 5**

BRANCH: IT, ECE

Blooms Taxonomy Levels:
Level 1 – Remembering, Level 2 – Understanding, Level 3 – Applying, Level 4 – Analyzing, Level 5 – Evaluating, Level 6 - Creating

Answer all questions.

S.No	Question	CO	BT Level
1a)	Explain the data science hype.	CO1	L2
1b)	Explain the features and uses of R programming and the RStudio IDE.	CO3	L2
2a)	Describe Drew Conways Venn diagram of data science.	CO1	L2
2b)	Explain the profile of a data scientist.	CO1	L2
3a)	Differentiate between population and sample	CO1	L2
3b)	Define bigdata and ambiguity in its terminology	CO1	L3
4	Explain the concept of N=ALL in the context of Big Data. Can it ensure unbiased analysis and accurate prediction. Refer to the journal paper below Symons, John, and Ramón Alvarado. "Can we trust Big Data? Applying philosophy of science to software." *Big Data & Society* 3.2 (2016): 2053951716664747. https://journals.sagepub.com/doi/pdf/10.1177/2053951716664747	CO1	L4
5a)	Explain the steps in exploratory data analysis.	CO2	L2
5b)	With a neat diagram explain the data science process	CO2	L2